

AIM:

To write and execute a Java program to demonstrate the commonly used methods of the Collection interface such as add(), addAll(), remove(), removeAll(), contains(), containsAll(), size(), isEmpty(), clear(), and iterator().

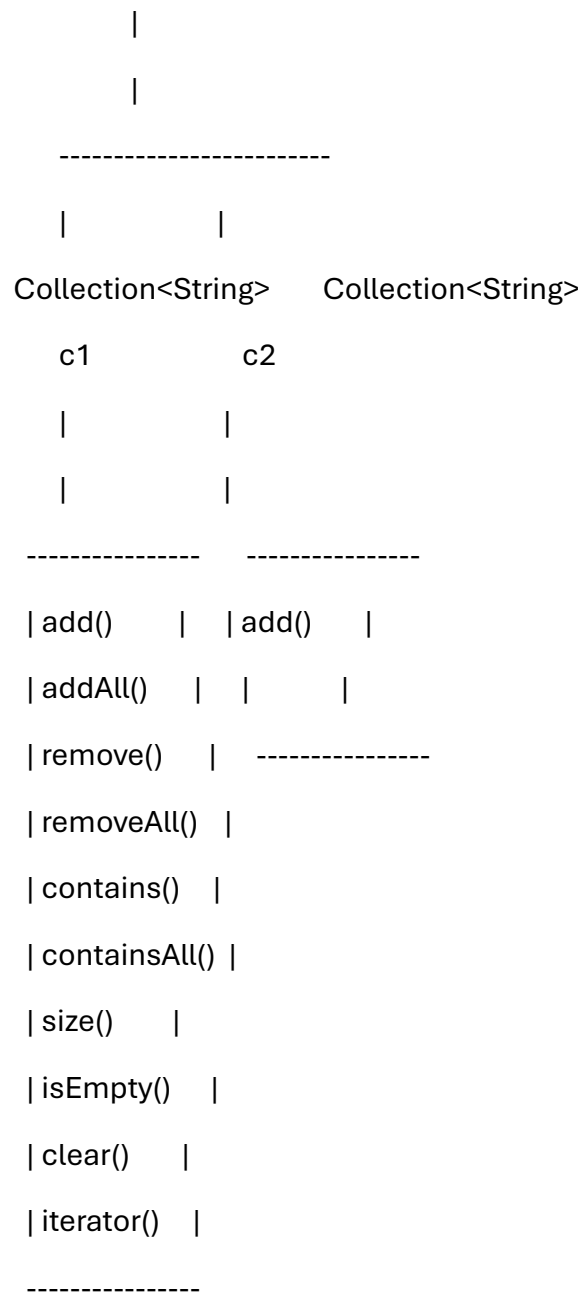
Algorithm:

Collection Interface Methods

1. **Start**
2. Import java.util.*.
3. Create a Collection<String> object using ArrayList.
4. Use add() to insert elements into the collection.
5. Create another collection and use addAll() to add all its elements.
6. Use remove() to remove a particular element.
7. Use removeAll() to remove all elements that are present in another collection.
8. Use contains() to check whether a particular element exists.
9. Use containsAll() to check whether all elements of another collection exist.
10. Use size() to find the number of elements.
11. Use isEmpty() to check whether the collection is empty.
12. Use iterator() to traverse and display all elements.
13. Use clear() to remove all elements from the collection.
14. Again use isEmpty() to verify that the collection is empty.
15. **Stop**

UML Diagram:

```
<<interface>>  
Collection<E>  
|  
| implements  
|  
<<class>>  
ArrayList<E>
```



Code:

```
package Observations;
```

```
import java.util.*;
```

```
public class CollectionMethods {
```

```
    public static void main(String[] args) {
```

```
        // Creating a Collection using ArrayList
```

```
Collection<String> c1 = new ArrayList<>();
```

```
// add(E e)
```

```
c1.add("Apple");
```

```
c1.add("Banana");
```

```
c1.add("Mango");
```

```
c1.add("Orange");
```

```
System.out.println("Collection: " + c1);
```

```
// addAll(Collection<? extends E> c)
```

```
Collection<String> c2 = new ArrayList<>();
```

```
c2.add("Grapes");
```

```
c2.add("Pineapple");
```

```
c1.addAll(c2);
```

```
System.out.println("After addAll(): " + c1);
```

```
// remove(Object o)
```

```
c1.remove("Banana");
```

```
System.out.println("After remove(): " + c1);
```

```
// removeAll(Collection<?> c)
```

```
c1.removeAll(c2);
```

```
System.out.println("After removeAll(): " + c1);
```

```
// contains(Object o)
```

```
System.out.println("Contains Mango? " + c1.contains("Mango"));
```

```

// containsAll(Collection<?> c)
Collection<String> c3 = new ArrayList<>();
c3.add("Apple");
c3.add("Mango");

System.out.println("Contains all c3 elements? "
    + c1.containsAll(c3));

// size()
System.out.println("Size of collection: " + c1.size());

// isEmpty()
System.out.println("Is collection empty? " + c1.isEmpty());

// iterator()
System.out.println("Elements using Iterator:");
Iterator<String> it = c1.iterator();

while (it.hasNext()) {
    System.out.println(it.next());
}

c1.clear();

System.out.println("After clear(): " + c1);

System.out.println("Is collection empty? " + c1.isEmpty());
}
}

```

Output:

```
bservations.CollectionMethods'  
Collection: [Apple, Banana, Mango, Orange]  
After addAll(): [Apple, Banana, Mango, Orange, Grapes, Pineapple]  
After remove(): [Apple, Mango, Orange, Grapes, Pineapple]  
After removeAll(): [Apple, Mango, Orange]  
Contains Mango? true  
Contains all c3 elements? true  
Size of collection: 3  
Is collection empty? false  
Elements using Iterator:  
Apple  
Mango  
Orange  
After clear(): []  
Is collection empty? true
```

Result:

Thus, the Java program to demonstrate the various methods of the Collection interface was successfully implemented and executed, and the required operations were performed successfully.